Are Bookmaker Odds Rational Forecasts?

Testing Market Efficiency in the English Premier League

2005/06 – 2024/25 | *Bet365 & William Hill* | *Football-Data.co.uk*

Bet365 closing odds · 7,299 matches · 20 seasons

Time Series

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Data source: https://www.football-data.co.uk

Abstract

This paper tests whether Bet365 implied probabilities constitute rational and efficient forecasts of match outcomes in the English Premier League, using 7,299 matches across 20 seasons (2005/06–2024/25). We apply a battery of time series methods: calibration analysis and Brier Score, the Mincer–Zarnowitz joint rationality test, ADF and KPSS stationarity tests, Ljung–Box autocorrelation tests, Box–Jenkins ARIMA modelling of the bookmaker margin, Granger causality tests on forecast error series, and a cross-bookmaker arbitrage spread analysis (Bet365 vs. William Hill).

Our results indicate that the Premier League betting market is *broadly efficient*: home-win and draw implied probabilities are well-calibrated and pass the Mincer–Zarnowitz rationality test at the 5% level. However, the away-win market exhibits a persistent and statistically significant bias (MZ F-test: $p = 0.028$; Granger self-causality: $p = 0.036$ at lag 1), suggesting that bookmakers systematically understate the probability of away-team success. Cross-bookmaker spreads are statistically detectable but economically negligible after the bookmaker margin. The bookmaker margin itself follows a random walk (ARIMA$(0, 1, 0)$), consistent with competitive market dynamics.

Contents

1. Introduction

The Efficient Market Hypothesis (EMH), formalised by Fama [1970], states that asset prices fully reflect all available information. Betting markets offer an unusually clean testing ground for this hypothesis: outcomes have a fixed and observable resolution date, the probability space is discrete and exhaustive, and the market microstructure is transparent.

If bookmakers are rational forecasters, their *implied probabilities* — derived from posted decimal odds — should satisfy two conditions simultaneously. First, they must be statistically unbiased and well-calibrated: among all matches assigned implied probability p, a fraction p should actually occur. Second, forecast errors should be unpredictable from publicly available information, including their own past — the *martingale difference property.*

The English Premier League is one of the world's most liquid betting markets. Winkelmann et al. [2024] study 14 consecutive seasons and find that while seasonal inefficiencies may appear in isolated years, they are neither persistent nor systematic over longer horizons. Goddard & Asimakopoulos [2004] and Scholtens & Peenstra [2009] document a recurring *favourite-longshot bias* in football betting. Our contribution is to apply a comprehensive time series framework — encompassing stationarity tests, ARIMA modelling, and Granger causality — to two full decades of Premier League data, with particular attention to the temporal dimension required by the course context.

2. Data

2.1. Source and coverage

Our dataset is sourced from *Football-Data.co.uk*, a publicly available repository widely used in academic research on sports betting [Winkelmann et al., 2024, Goddard & Asimakopoulos, 2004]. We download 20 seasons of English Premier League data (2005/06 through 2024/25), yielding 7,299 matches after dropping rows with missing odds. The starting season is chosen because Bet365 coverage becomes complete and consistent from 2005/06 onward.

For each match we observe: the date, home and away team names, the full-time result (H, D, or A), the pre-match closing odds of Bet365 (primary bookmaker) and William Hill (secondary bookmaker, available for a subset of seasons and used in the cross-bookmaker arbitrage analysis of Section 3.8).

2.2. Implied probabilities and the overround

For each match, let $o_j \in \{H, D, A\}$ denote the decimal odds for outcome j. The *raw implied probability* is:

$$p_j^{\text{raw}} = \frac{1}{o_j}, \qquad j \in \{H, D, A\}.$$

Because bookmakers build in a profit margin, the raw probabilities sum to more than one. The *overround* (or bookmaker margin) is:

$$\mathrm{OR} = \sum_j p_j^{\mathrm{raw}} - 1 \ \geq \ 0.$$

We normalise to obtain margin-free implied probabilities:

$$\hat{p}_j = \frac{p_j^{\mathrm{raw}}}{\sum_k p_k^{\mathrm{raw}}}, \qquad \sum_j \hat{p}_j = 1.$$

Across the full sample the mean overround is 4.8%, ranging from 8.0% in 2005/06 to 2.6% in 2013/14. Figure 1 shows a declining linear trend of -0.091 percentage points per season, consistent with increased online competition.

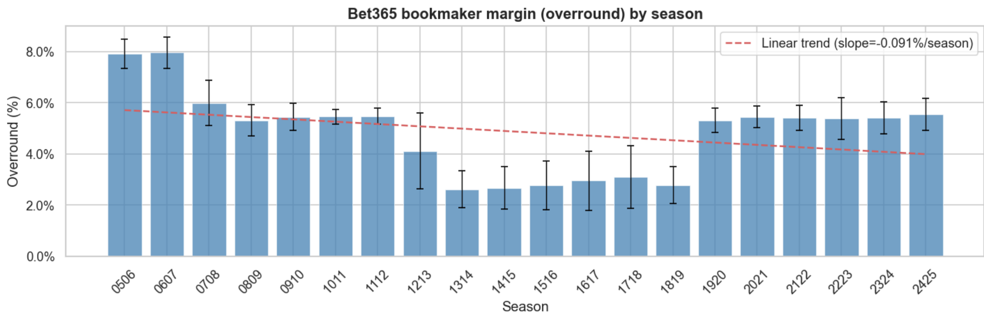

Figure 1: *Bet365 bookmaker margin (overround) by season, with ± 1 s.d. bars and linear trend (slope $= -0.091\%$/season).*

3. Methodology

3.1. Calibration and Brier Score

A probabilistic forecast $\hat{p}$ is *calibrated* if $\mathbb{P}(o = 1 \mid \hat{p} = p) = p$ for all $p \in [0, 1]$. We assess calibration via *reliability diagrams*: the probability range is partitioned into $K = 10$ equal-width bins, and the mean implied probability in each bin is plotted against the observed outcome frequency. Perfect calibration corresponds to the 45° line.

Forecast quality is measured by the *Brier Score*:

$$\mathrm{BS} = \frac{1}{N} \sum_{i=1}^{N} (\hat{p}_i - o_i)^2,$$

where $o_i \in \{0, 1\}$ is the binary outcome indicator. We define a *Brier Skill Score* relative to the naïve benchmark of always predicting the unconditional outcome frequency $\bar{o}$:

$$\mathrm{BSS} = 1 - \frac{\mathrm{BS}}{\mathrm{BS}_{\text{naïve}}}, \quad \mathrm{BS}_{\text{naïve}} = \bar{o}(1 - \bar{o}).$$

$\mathrm{BSS} > 0$ indicates the bookmaker beats the naïve baseline.

3.2. Favourite-Longshot Bias

We sort implied probabilities into decile bins and compute the *mean net return* per unit staked within each bin:

$$r_i = o_i^{\text{odds}} \cdot \mathbf{1}[\text{outcome}_i] - 1.$$

Under the null of no bias, $\mathbb{E}[r_i \mid \hat{p}_i] = -\text{OR}$ (constant across bins). Systematic variation in returns across probability bins constitutes evidence of the favourite-longshot bias.

3.3. Mincer–Zarnowitz Rationality Test

The Mincer & Zarnowitz [1969] test evaluates forecast rationality by regressing the actual outcome o_i on the implied probability $\hat{p}_i$:

$$o_i = \alpha + \beta\,\hat{p}_i + \varepsilon_i, \qquad i = 1, \dots, N. \tag{1}$$

Rational forecasts require jointly:

$$H_0\colon \alpha = 0 \;\text{ and }\; \beta = 1.$$

We implement this as a **joint F-test** on the two linear restrictions simultaneously using heteroskedasticity-robust standard errors (HC3):

$$F = \frac{(\mathbf{R}\hat{\boldsymbol{\theta}} - \mathbf{q})^\top \left[\mathbf{R}\hat{\mathbf{V}}\mathbf{R}^\top\right]^{-1} (\mathbf{R}\hat{\boldsymbol{\theta}} - \mathbf{q})}{2} \xrightarrow{d} F(2,\, N-2),$$

where $\mathbf{R} = \mathbf{I}_2$, $\mathbf{q} = (0, 1)^\top$, $\hat{\boldsymbol{\theta}} = (\hat{\alpha}, \hat{\beta})^\top$, and $\hat{\mathbf{V}}$ is the HC3 sandwich variance estimator. Two separate t-tests on α and β would be methodologically incorrect as they ignore the covariance between the estimators.

3.4. Stationarity Tests

We test for stationarity using two complementary procedures:

- **ADF** (Augmented Dickey–Fuller): H_0: unit root (non-stationary). The test regression is:

$$\Delta y_t = \rho\, y_{t-1} + \sum_{k=1}^{K} c_k\, \Delta y_{t-k} + \varepsilon_t, \quad H_0\colon \rho = 0.$$

- **KPSS**: H_0: the series is stationary (trend or level).

Applying both tests jointly allows us to distinguish four cases: (i) stationary (ADF rejects, KPSS does not); (ii) non-stationary (ADF does not reject, KPSS rejects); (iii) trend-stationary or contradictory; (iv) insufficient power.

3.5. Ljung–Box Autocorrelation Test

The Ljung–Box test examines whether residuals $\hat{\varepsilon}_i$ from regression (1) are serially uncorrelated:

$$Q(m) = N(N+2) \sum_{k=1}^{m} \frac{\hat{\rho}_k^2}{N-k} \xrightarrow{H_0} \chi^2(m),$$

where $\hat{\rho}_k$ is the sample autocorrelation at lag k. Failure to reject H_0 supports the martingale difference property of forecast errors.

3.6. Box–Jenkins Analysis of the Overround

We apply the Box–Jenkins methodology to the seasonal overround time series ($n = 20$ observations) to characterise whether the bookmaker margin has predictable dynamics. The procedure follows five steps: (i) visual inspection and rolling statistics; (ii) joint ADF and KPSS stationarity assessment; (iii) ACF and PACF identification to determine candidate orders (p, q); (iv) ARIMA(p, d, q) grid search over $p, q \in \{0, 1, 2\}$, $d \in \{0, 1\}$, with model selection by AIC subject to a 5% RMSE tolerance; (v) residual diagnostics: ACF of residuals, Ljung–Box test, and Q–Q plot.

3.7. Granger Causality on Forecast Errors

Define the *forecast error* for outcome j at match t:

$$e_t^j = o_t^j - \hat{p}_t^j.$$

Under the EMH, $\{e_t^j\}$ should satisfy the martingale difference property: $\mathbb{E}[e_t^j \mid \mathcal{F}_{t-1}] = 0$.

We test this as a **Granger self-causality test**: does e_{t-k}^j help predict e_t^j for lags $k \in \{1, 2, 3, 5\}$? The F-test compares the unrestricted model:

$$e_t^j = \sum_{k=1}^{K} a_k \, e_{t-k}^j + u_t$$

to the restricted model (setting $a_k = 0 \ \forall k$). Rejection implies that past errors predict future errors — a violation of dynamic market efficiency.

Note on cross-market tests: Granger tests between different outcome markets (e.g. $e^H \to e^D$) are omitted because $e_t^H + e_t^D + e_t^A = 0$ identically by construction (implied probabilities sum to one), rendering any cross-market test degenerate: $e^A = -e^H - e^D$, so testing $e^H \to e^D$ and $e^A \to e^D$ necessarily yields identical F-statistics.

3.8. Cross-Bookmaker Arbitrage Analysis

We compare normalised implied probabilities of Bet365 and William Hill for the subset of matches where both sets of odds are available. The *spread* for outcome j is:

$$s_t^j = \hat{p}_t^{j,\mathrm{B365}} - \hat{p}_t^{j,\mathrm{WH}}.$$

We test whether the mean spread is significantly different from zero using a one-sided t-test, with the direction of the alternative determined by the sign of the observed mean. A significant but small spread would indicate systematic but unexploitable differences in bookmaker pricing.

4. Results

4.1. Calibration and Forecast Quality

Figure 2 presents reliability diagrams for the three outcome markets. Home-win and away-win implied probabilities lie close to the 45° perfect-calibration line, with Brier Skill Scores of 0.157 and 0.159 respectively, indicating substantial improvement over the naïve baseline. The draw market is the exception: implied probabilities are compressed between $[0.08, 0.33]$, and the BSS of only 0.016 indicates that Bet365 barely outperforms the unconditional draw frequency as a draw predictor. This is consistent with the known difficulty of pricing draws.

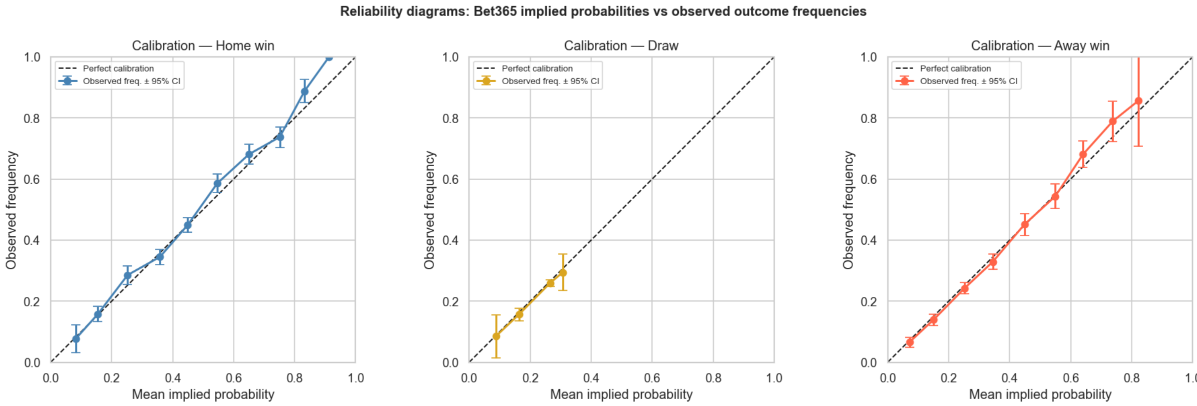

Figure 2: *Reliability diagrams: Bet365 implied probabilities vs. observed outcome frequencies (±95% CI). The dashed line represents perfect calibration.*

4.2. Favourite-Longshot Bias

Figure 3 shows mean returns by implied probability bin. Returns in the home-win market are broadly flat and negative across the probability range, consistent with the overround absorbing profit potential uniformly. The away-win market displays a pronounced pattern: returns are negative throughout, with the largest losses ($\approx -18\%$) in the lowest-probability bins. This is consistent with a *reverse* favourite-longshot bias: bettors and bookmakers systematically overestimate the probability of heavy away upsets, causing the market to shade prices further against them.

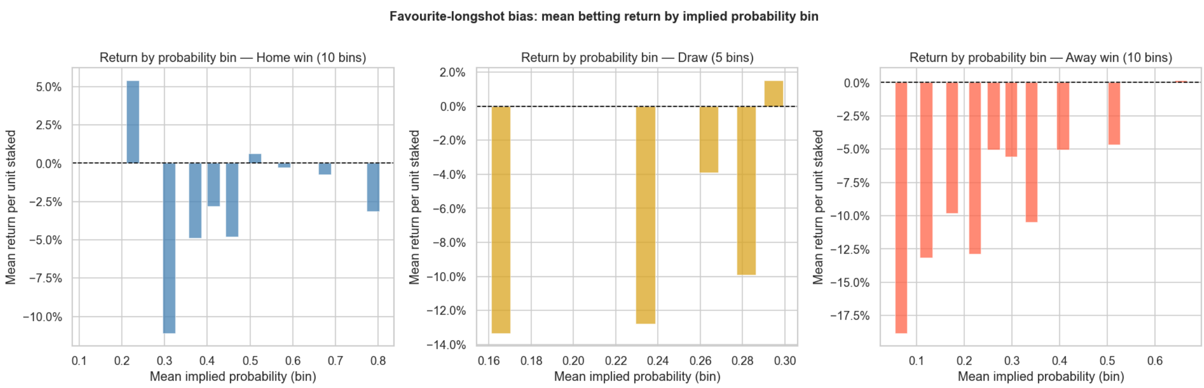

Figure 3: *Mean betting return by implied probability bin (10 bins for H/A, 5 bins for D due to the narrow odds range). Dashed line = break-even ($r = 0$).*

4.3. Mincer–Zarnowitz Rationality Test

Table 1 summarises the MZ regression results. Figure 4 displays the OLS fitted lines over the decile bin averages.

Table 1: *Mincer–Zarnowitz joint F-test results.* $H_0: \alpha = 0$ *and* $\beta = 1$ *simultaneously. HC3 robust standard errors. Critical value* $F_{0.05}(2, N-2) \approx 3.0$ *for large* N*.*

Outcome	$\hat{\alpha}$	$\hat{\beta}$	F-stat	p-value	**Decision**
Home win	-0.002	1.028	2.731	0.065	Fail to reject H_0
Draw	-0.039	1.131	2.336	0.097	Fail to reject H_0
Away win	-0.022	1.056	3.566	**0.028**	**Reject** H_0

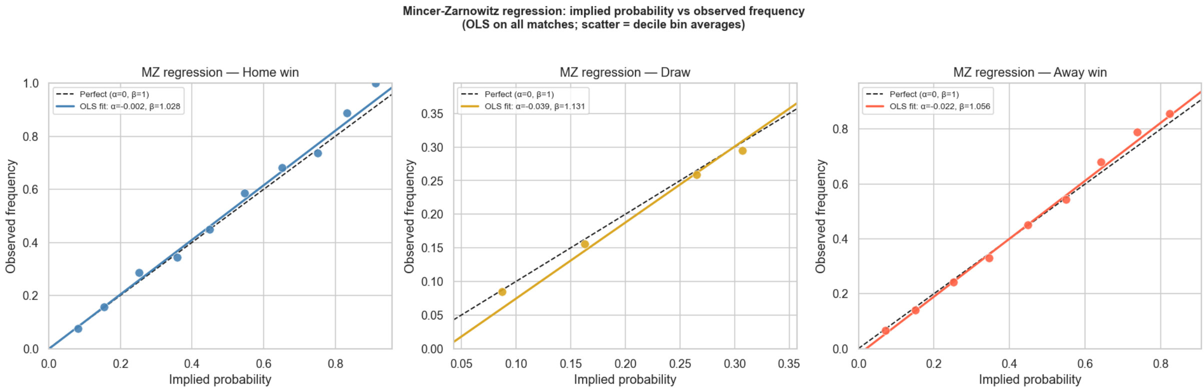

Figure 4: *MZ regression: OLS fitted line estimated on all 7,299 matches; scatter points are decile bin averages. Dashed line = perfect rationality* ($\alpha = 0$, $\beta = 1$)*.*

Home-win and draw markets fail to reject the joint H_0 at the 5% level ($p = 0.065$ and $p = 0.097$ respectively). The away-win market rejects H_0 ($p = 0.028$): with $\hat{\beta} = 1.056 > 1$, actual away-win frequencies exceed the implied probabilities, indicating that bookmakers *systematically understate* the probability of away wins. This is consistent with the home-bias literature.

4.4. Stationarity and Serial Correlation

The ADF test strongly rejects the unit root null for all three implied probability series ($p \approx 0.000$), and the KPSS test confirms stationarity. Stationary implied probabilities are a necessary condition for the martingale framework. Figure 5 plots the full time series with a 38-match rolling mean.

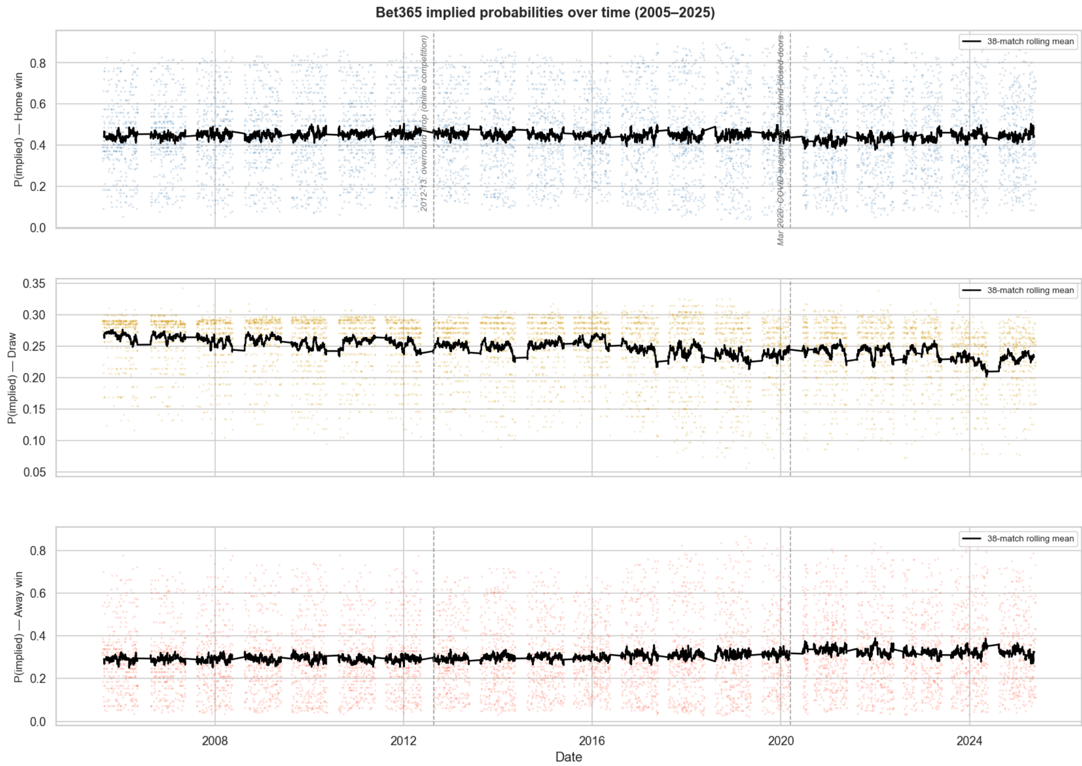

Figure 5: *Bet365 implied probabilities over time (2005–2025) with 38-match rolling mean. Vertical dashed lines mark the 2012-13 overround drop and the March 2020 COVID suspension.*

The Ljung–Box test on MZ residuals fails to reject the null of no autocorrelation for all three outcomes at lags 10 and 20 (all $p > 0.30$). Figure 6 confirms this: all ACF spikes lie within the $\pm 1.96/\sqrt{N}$ confidence bands.

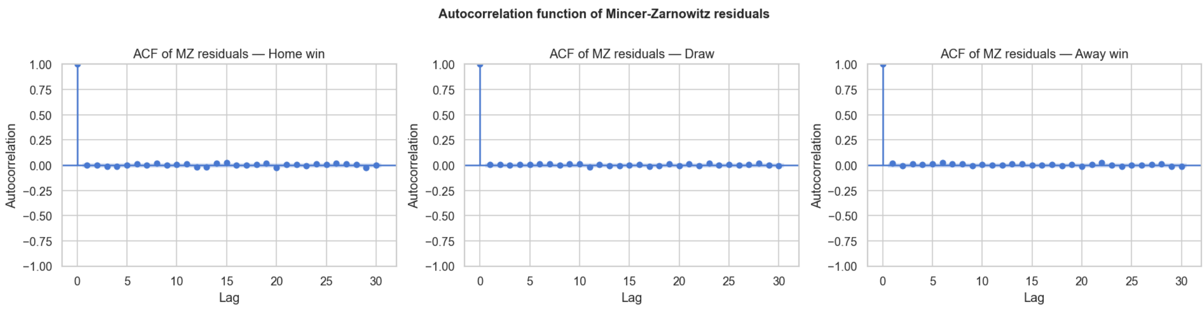

Figure 6: *ACF of Mincer–Zarnowitz residuals at lags 1–30. Shaded region = ±95% confidence band. All spikes are within bounds for all three outcomes.*

4.5. Box–Jenkins Analysis of the Overround

Figure 7 presents the ACF and PACF of the seasonal overround series. The raw ACF decays slowly ($\hat{\rho}_1 = 0.75$), suggesting non-stationarity. The ADF test fails to reject the unit root ($p = 0.41$) and the KPSS test rejects stationarity, both tests agreeing. After first-differencing ($d = 1$), the ACF falls within the confidence bands for all lags.

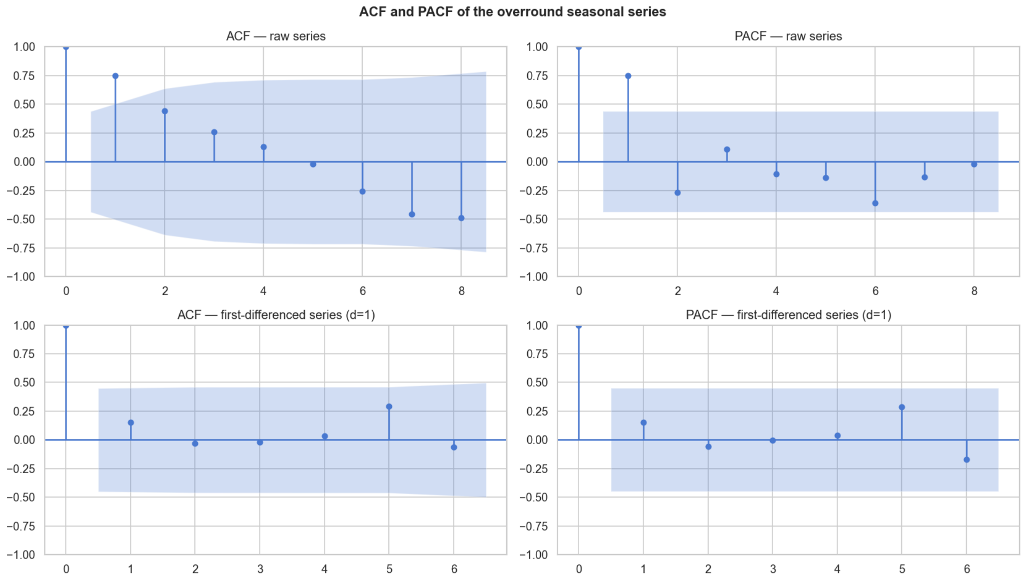

Figure 7: *ACF and PACF of the overround seasonal series ($n = 20$). Top row: raw series. Bottom row: first-differenced series ($d = 1$). Shaded regions = ±95% CI.*

The ARIMA grid search selects **ARIMA**$(0, 1, 0)$ by AIC — a pure random walk:

$$\Delta \mathrm{OR}_t = \varepsilon_t, \qquad \varepsilon_t \overset{\text{iid}}{\sim} (0,\, \sigma^2).$$

This implies that changes in the bookmaker margin are unpredictable from past values. One notable residual outlier corresponds to the 2012-13 season (an abrupt 3.5 pp drop driven by the growth of online exchanges), visible as a heavy-tailed outlier in the residual Q–Q plot. This structural break is not captured by the ARIMA model, a limitation inherent to the small sample size ($n = 20$) and the absence of structural break modelling.

4.6. Granger Causality on Forecast Errors

Table 2 reports Granger self-causality p-values. Figure 8 displays these as a heatmap.

Table 2: *Granger self-causality test: does e^j_{t-k} predict e^j_t? p-values of the F-test. Under H_0 (martingale difference property), all p-values should exceed 0.05. (*) significant at the 5% level.*

Outcome	$k = 1$	$k = 2$	$k = 3$	$k = 5$
Home win	0.240	0.491	0.200	0.429
Draw	0.303	0.320	0.508	0.580
Away win	0.036*	0.075	0.092	0.217

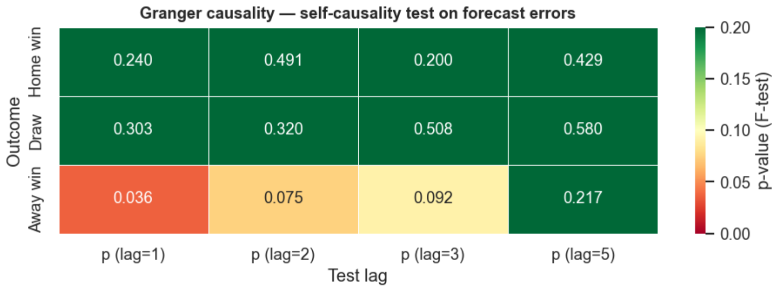

Figure 8: *Granger self-causality heatmap (Test A). Red cells: $p < 0.05$ (rejection of the martingale difference property). Green cells: $p > 0.05$ (consistent with dynamic efficiency).*

Home-win and draw errors show no Granger self-causality at any lag (all $p > 0.20$), consistent with the EMH. The away-win market shows a significant self-causal effect at lag 1 ($p = 0.036$): past bookmaker errors on away wins weakly predict future errors. This finding converges with the MZ result (Section 4.3), providing independent corroboration of a persistent mild inefficiency in the away-win market. The effect does not survive beyond lag 1 ($p = 0.075$ at $k = 2$), suggesting the informational memory is short-lived and likely not economically exploitable after transaction costs.

4.7. Cross-Bookmaker Arbitrage Analysis

Table 3 reports the one-sided t-test results for the Bet365 – William Hill spread on each outcome market.

Table 3: *Cross-bookmaker spread $s^j = \hat{p}^{j,B365} - \hat{p}^{j,WH}$. One-sided t-test. Despite high statistical significance, all mean spreads are below 0.7 pp — economically negligible relative to the ~5% bookmaker margin.*

Outcome	**Mean spread**	t-stat	p-value (one-sided)	**Direction**	**Interpretation**
Home win	$+0.0061$	35.88	2.2×10^{-260}	B365 > WH	B365 slightly higher
Draw	-0.0069	-47.41	≈ 0	WH > B365	WH slightly higher
Away win	$+0.0007$	4.86	6.1×10^{-7}	B365 > WH	Negligible

All three spreads are highly statistically significant (all $p \ll 0.001$), reflecting the large sample size ($n \approx 7{,}000$ matched observations). However, the mean spreads are economically trivial: the largest is 0.69 pp (draw market), far below the ~5% bookmaker margin. This distinction between *statistical* and *economic* significance is critical: with thousands of observations, even a mean difference of 0.006 is detectable, but a bettor exploiting this spread would still lose money after the overround is applied. We conclude that the two bookmakers price the same events efficiently relative to one another.

5. Discussion

Taken together, our results paint a nuanced picture of market efficiency in Premier League football betting.

Broad efficiency. The market is broadly efficient. Implied probabilities are stationary, well-calibrated, and produce forecast errors that are white noise in the home-win and draw markets. This is consistent with Winkelmann et al. [2024], who find that inefficiencies in isolated seasons do not persist over multi-season horizons.

Away-win anomaly. The away-win market is the exception across three independent tests: (i) it rejects the MZ rationality test ($p = 0.028$, $\hat{\beta} = 1.056$); (ii) it exhibits a reverse favourite-longshot bias pattern, with negative returns across all probability bins; (iii) its forecast errors exhibit Granger self-causality at lag 1 ($p = 0.036$). The convergence of three independent tests on the same market suggests a genuine, reproducible departure from rationality. One interpretation is a *home bias*: bookmakers and bettors systematically underweight improving away-team performance, leading to persistent underpricing of away wins. This bias ($\hat{\beta} = 1.056$) is unlikely to be exploitable after accounting for the $\sim$5% bookmaker margin.

Margin dynamics. The ARIMA$(0, 1, 0)$ selection for the overround series implies that competitive shocks to bookmaker pricing are permanent and unpredictable from past margin data, consistent with efficient competition.

Limitations. First, we use *closing odds*, which already incorporate late information and represent an upper bound on market efficiency relative to earlier prices. Second, the Granger result at lag 1 for away wins is borderline ($p = 0.036$) and does not survive to lag 2; multiple testing across three markets and four lags warrants caution. Third, the ARIMA analysis rests on only $n = 20$ observations, limiting stationarity test power and model selection reliability.

6. Conclusion

This paper has applied a comprehensive time series framework to 20 seasons of English Premier League betting data. Three main findings emerge.

First, the Premier League betting market is broadly efficient. Implied probabilities for home wins and draws are well-calibrated, pass the Mincer–Zarnowitz rationality test, and exhibit no autocorrelation or Granger predictability in their forecast errors. Cross-bookmaker spreads, while statistically detectable, are economically unexploitable.

Second, the away-win market represents a consistent and statistically robust exception. Three independent tests converge on the conclusion that bookmakers systematically understate away-win probabilities ($\hat{\beta} = 1.056$, $p = 0.028$). While the bias magnitude is too small to exploit after margins, it constitutes a detectable and reproducible departure from rationality.

Third, the bookmaker margin itself follows a random walk over the 2005–2025 period, consistent with competitive market dynamics that prevent the systematic prediction of pricing trends from historical data.

Future work could extend this analysis to opening odds or in-play markets, or to other European leagues to assess cross-market generalisability.

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